

Pop Quiz

Date: _____

Questions

1. Define the following in your own words:

(a) Probability

THE CHANCE OF SOMETHING OCCURRING
SHOWN AS A NUMBER 0-1

(b) Experiment

SOME CONTROLLED ACTION WHERE THE OUTCOME
IS DETERMINED BY CHANCE

(c) Sample Space

THE TOTAL SPACE OF OCCURRENCES INSIDE
THE EXPERIMENT

(d) Event

A SUBSET OF THE SAMPLE SPACE

2. ^a Explain what it means for an event to have a probability $P = 0.001$.^b Find that event's complement and explain what that probability means.

(a) IT'S UNLIKELY BUT NOT IMPOSSIBLE

(b) LIKELY BUT POSSIBLE FOR IT NOT TO OCCUR

3. Let A and B be independent events with probabilities, $P(A) = p_1$, $P(B) = p_2$.(a) Find $P(A \cup B)$.

$$*P(A|B) = P(A) \quad P(B|A) = P(B)$$

(b) Find the probability that B occurs n times in sequence, given that we've observed event A .

$$\textcircled{a} \quad \frac{P(A \cap B)}{P(B)} = P(A) \quad P(A \cap B) = P(A)P(B) \quad P(A \cup B) = P(A) + P(B) - P(A)P(B)$$

$$p_1 + p_2 - p_1 p_2$$

$$\textcircled{b} \quad P(B|A)^n = P(B)^n = p_2^n$$